

Protective ventilation to prevent secondary brain injury – PReVENT2BI study

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Background: Application of protective mechanical ventilation is recommended in patients with severe traumatic brain injury^{1,2} due to VILI predisposition as a result of neuroinflammatory response and the adrenergic cascade³ and increased incidence of ARDS (15–20%).^{4,5} However, low tidal volume strategies and higher levels of PEEP could lead to hypercapnia and decreased cerebral venous drainage, resulting in higher incidence of secondary brain injury.^{6,7}

Objective: To assess the impact of the changes in airway pressures during mechanical ventilation on cerebral perfusion pressure (CPP)

Methods and analysis: A retrospective study evaluating mechanically ventilated patients with traumatic brain injury (TBI) from the AmsterdamUMC ICU Database was performed. 7189 mechanically ventilated patients with PEEP and central venous pressure (CVP) measurements after excluding factors that compromise the accuracy of CVP measurement, were split into two groups: No TBI group - 5716 patients and TBI group - 1473 patients with ICP measurements, of whom 279 had continuous measurements of CVP and ICP (33138 instances) and mechanical ventilation parameters.

For each patient, parameters for assessment of protective mechanical ventilation (PEEP, DP, Pplat, I:E ratio, FiO₂, pCO₂, pH, disconnection/aspiration per day), mechanical power (RR, V_T, Paw mean) and adequacy of mechanical ventilation (pO₂, pCO₂, pH, P/F ratio, A-a gradient, O₂ER), were derived. These data sets narrowed an optimal permutation of input features that an approximator could rely on to find non-linear patterns.

XGBoost is an ensemble supervised-learning method based on the idea of gradient boosting with decision trees. It minimizes a global objective function that combines a loss function (such as MSE, MAE, Huber) with a regularization term to penalize model complexity. The loss is approximated using a second-order Taylor expansion, conveying more accurate and stable decisions during tree construction. Node splits are chosen by maximizing the gain, which measures loss reduction while accounting for regularization. Each new tree is added to the prediction with a learning-rate factor for gradual, controlled improvement. By combining gradient-based optimization, second-order information, stochastic sampling, and strong regularization, XGBoost achieves high accuracy and strong generalization. As an initial modeling effort to mitigate risk of overfitting, a four based ensemble learners: XGBoost, LightGBM, CatBoost, and Random Forest, proved to sufficiently combine their outputs as a

two-stage predictor that strengthened generalization with new samples, which and lend an insight for a simplified model once each learner contribution wise analysed.

Results and Discussion: A total of 7189 patients were analyzed. The XGBoost approximator demonstrated capacity to capture underlying patterns effectively. Upon validation with unseen test data, the model yielded sufficient accuracy, indicating its generalizability supporting its predictive capability (Fig. 1).

Studying the samples, we've noticed that not alone airway pressure has influence on venous pressure levels, yet also a small increase in PEEP contributes to increase in CVP (Fig.2).

Conclusions: Our study suggests that slight adjustment in PEEP can influence CVP to have clinically significant effects on brain injury. To aid personalization of lung protective ventilation an approximator was trained on clinical data to forecast potential CVP and ICP pressure levels.

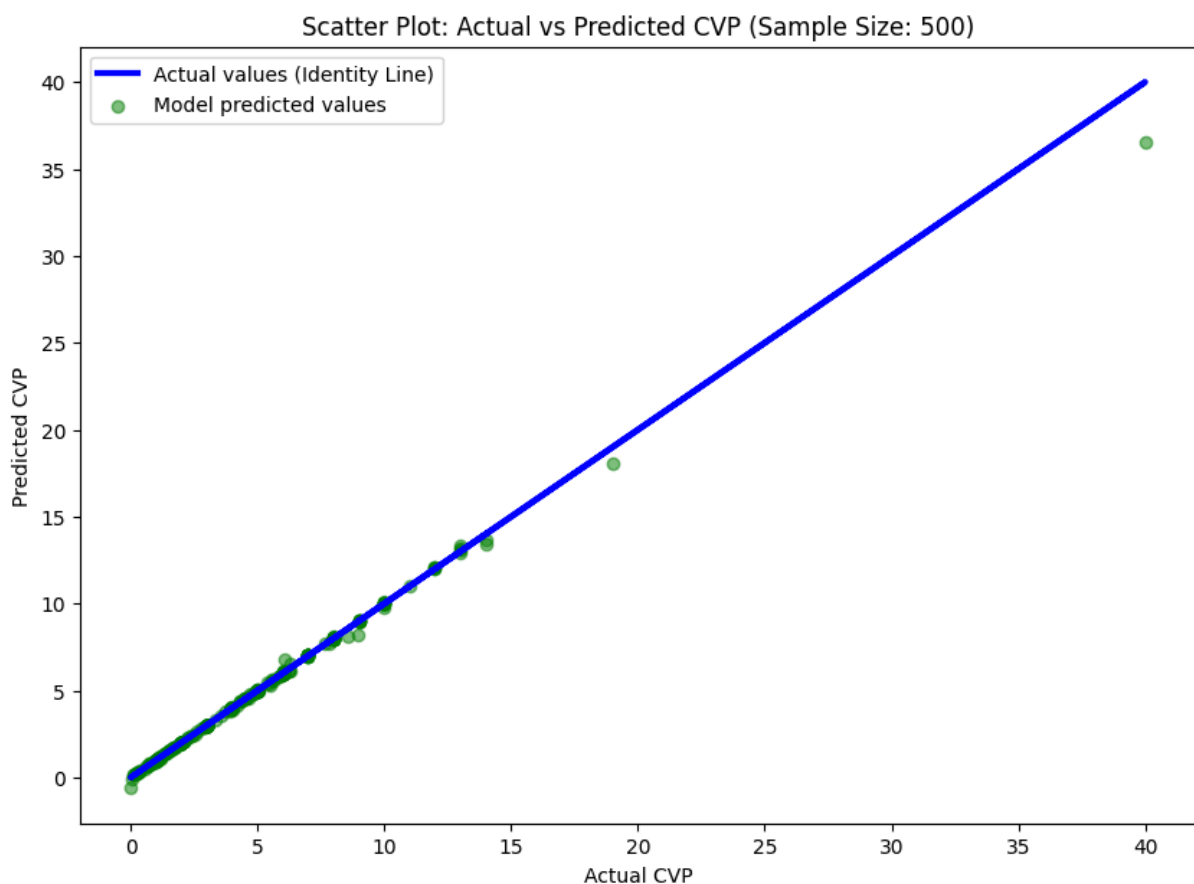


Fig. 1. Predicted CVP values obtained by the XGBoost regressor using the hist-based tree method and an 80/20 train-test split. The model employed a deep architecture, L1 and L2 regularization to mitigate overfitting, and early stopping using a squared error objective. Feature importance was assessed via Gain scores to identify the most influential clinical variables. The model is mostly influenced by input values of Δ PEEP and previous measurement of CVP. *Modelling results:* 1) On training data: Mean Absolute Error (MAE): 0.0309, Mean Squared Error (MSE): 0.0647, Root Mean Squared Error (RMSE): 0.2545, and R^2 Score: 0.9930; 2) On the testset: RMSE: 0.2068, Test MAE: 0.0297, Test R^2 : 0.9954

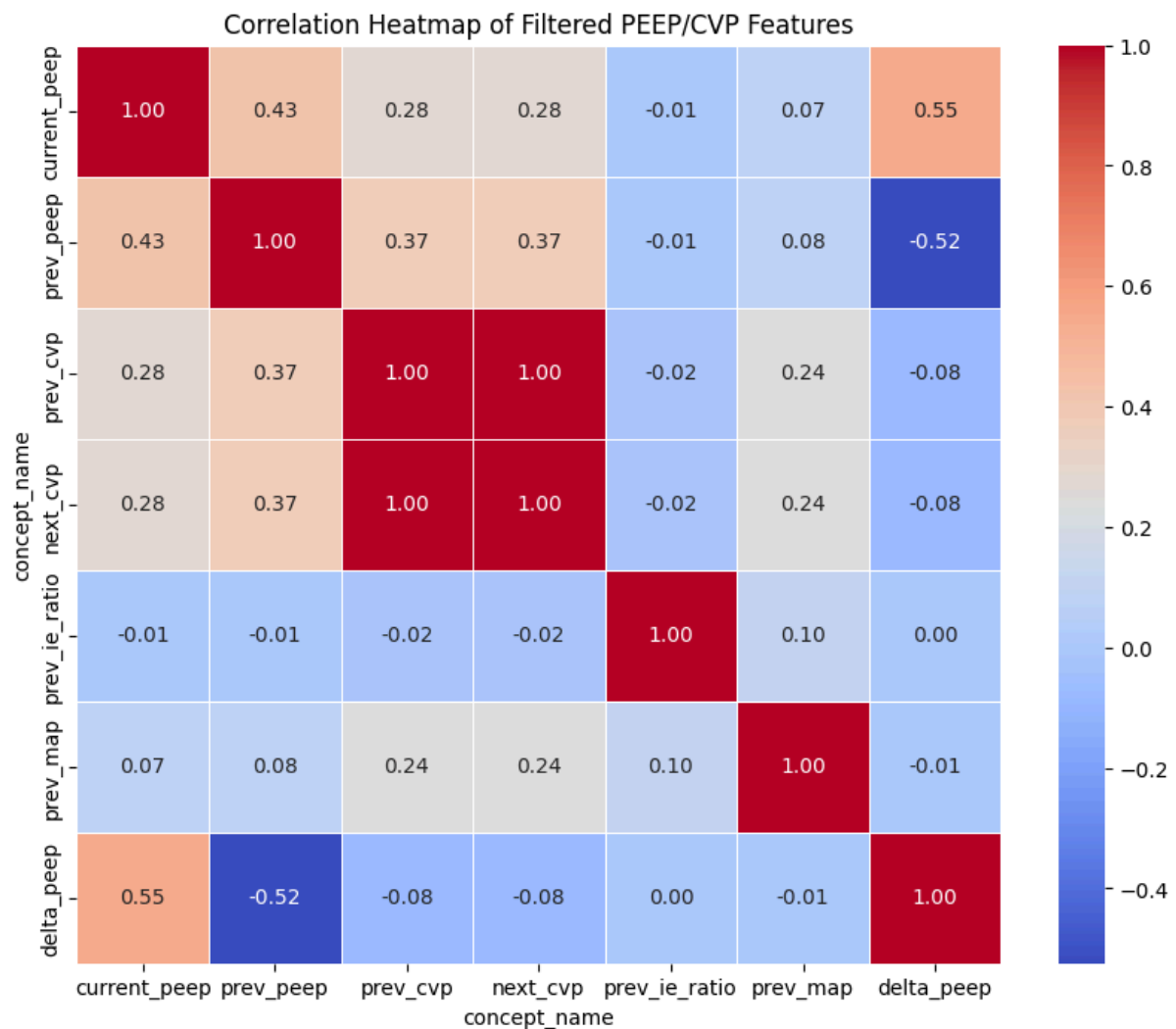


Fig 2. Studies the correlative effects between small increases in PEEP, airway pressure and CVP.

References:

1. Picetti E, Pelosi P, Taccone FS, Citerio G, Mancebo J, Robba C; ESICM NIC/ARF sections. VENTILatOry strategies in patients with severe traumatic brain injury: the VENTILO Survey of the European Society of Intensive Care Medicine (ESICM). *Crit Care* 2020;24:158.
2. Asehnoune K, Mrozek S, Perrigault PF, Seguin P, Dahyot-Fizelier C, Lasocki S, et al.; BI-VILI Study Group. A multi-faceted strategy to reduce ventilation-associated mortality in brain-injured patients. The BI-VILI project: a nationwide quality improvement project. *Intensive Care Med* 2017;43:957–970.
3. Mascia L, Pasero D, Shutsky AS, Arguis MJ, Berardino M, Grasso S, et al. Effect of a lung protective strategy for organ donors on eligibility and availability of lungs for transplantation: a randomized controlled trial. *JAMA* 2010;304:2620–2627.
4. Holland MC, Mackersie RC, Morabito D, Campbell AR, Kivett VA, Patel R, et al. The development of acute lung injury is associated with worse neurologic outcome in patients with severe traumatic brain injury. *J Trauma* 2003;55:106–111.
5. Mascia L, Sakr Y, Pasero D, Payen D, Reinhart K, Vincent JL; Sepsis Occurrence in Acutely Ill Patients (SOAP) Investigators. Extracranial complications in patients with acute brain injury: a post-hoc analysis of the SOAP study. *Intensive Care Med* 2008;34:720–727.
6. Citerio G, Robba C, Rebora P, Petrosino M, Rossi E, Malgeri L, et al.; Center-TBI Participants and Investigators. Management of arterial partial pressure of carbon dioxide in the first week after traumatic brain injury: results from the CENTER-TBI study. *Intensive Care Med* 2021; 47:961–973.
7. Boone MD, Jinadasa SP, Mueller A, Shaefi S, Kasper EM, Hanafy KA, et al. The effect of positive end-expiratory pressure on intracranial pressure and cerebral hemodynamics. *Neurocrit Care* 2017;26: 174–181.